

EXPERIMENT 1(a)

Question 1. Write a program to print addition of 2 matrices in java.

```
import java.util.Scanner;
public class MatricesAdd{
    public static void main(String[] args){
        System.out.println("The program is coded by Karunya Singh
Kanyal, 2400320100580");
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the number of rows:");
        int rows=scanner.nextInt();
        System.out.print("Enter the number of columns:");
        int cols=scanner.nextInt();
        int[][] matrix1=new int[rows][cols];
        int[][] matrix2=new int[rows][cols];
        System.out.println("Enter the elements of the first matrix:");
        for(int i=0;i<rows;i++){
            for(int j=0;j<cols;j++){
                matrix1[i][j]=scanner.nextInt();
            }
        }
        System.out.println("Enter the elements of the second
matrix:");
        for(int i=0;i<rows;i++){
            for(int j=0;j<cols;j++){
                matrix2[i][j]=scanner.nextInt();
            }
        }
        int[][] sum=new int[rows][cols];
        for(int i=0;i<rows;i++){
            for(int j=0;j<cols;j++){
                sum[i][j]=matrix1[i][j]+matrix2[i][j];
            }
        }
        System.out.println("The sum of the two matrices is:");
        for(int i=0;i<rows;i++){
            for(int j=0;j<cols;j++){
                System.out.print(sum[i][j]+" ");
            }
            System.out.println();
        }
    }
}
```

Output –

The program is coded by Karunya Singh Kanyal, 2400320100580

Enter the number of rows:3

Enter the number of columns:3

Enter the elements of the first matrix:

1 1 1

2 2 2

3 3 3

Enter the elements of the second matrix:

4 4 4

5 5 5

6 6 6

The sum of the two matrices is:

5 5 5

7 7 7

9 9 9

1(b)

Question 2. Write a program in java which creates the variable size array (Jagged Array) and print all the values using loop statement.

```
public class jaggedarray {  
    public static void main(String[] args) {  
        System.out.println("The program is coded by Karunya Singh Kanyal, 2400320100580");  
  
        int[][] jaggedArray = new int[3][];  
        jaggedArray[0] = new int[2];  
        jaggedArray[1] = new int[3];  
        jaggedArray[2] = new int[4];  
        int count = 1;  
        for(int i = 0; i < jaggedArray.length; i++) {  
            for(int j = 0; j < jaggedArray[i].length; j++) {  
                jaggedArray[i][j] = count++;  
            }  
        }  
  
        System.out.println("The jagged array is:");  
        for(int i = 0; i < jaggedArray.length; i++) {  
            for(int j = 0; j < jaggedArray[i].length; j++) {  
                System.out.print(jaggedArray[i][j] + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output -

The program is coded by Karunya Singh Kanyal, 2400320100580

The jagged array is:

1 2

3 4 5

6 7 8 9

EXPERIMENT 2(a)

Question 3. WAP that takes input from user through command line argument and then prints whether a number is prime or no

```
import java.util.Scanner;
public class prime {
    public static void main(String[] args) {
        System.out.println("The program is coded by Karunya Singh
Kanyal, 2400320100580");

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        boolean isPrime = true;

        if(num <= 1) {
            isPrime = false;
        } else {
            for(int i = 2; i <= Math.sqrt(num); i++) {
                if(num % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if(isPrime) {
            System.out.println(num + " is a prime number");
        } else {
            System.out.println(num + " is not a prime number");
        }

        sc.close();
    }
}
```

Output –

```
The program is coded by Karunya Singh Kanyal, 2400320100580
Enter a number: 29
29 is a prime number
```

EXPERIMENT 2(b)

Question 4. Write a program to enter number through command line and check whether it is palindrome or not.

```
import java.util.Scanner;
public class palindrome {
    public static void main(String[] args) {
        System.out.println("The program is coded by Karunya Singh
Kanyal, 2400320100580");

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String reversed = "";

        for(int i = str.length() - 1; i >= 0; i--) {
            reversed += str.charAt(i);
        }

        if(str.equals(reversed)) {
            System.out.println("The string is a palindrome");
        } else {
            System.out.println("The string is not a palindrome");
        }

        sc.close();
    }
}
```

Output –

```
The program is coded by Karunya Singh Kanyal, 2400320100580
Enter a string: madam
The string is a palindrome
```

EXPERIMENT 3(a)

Question 5. Write a Java program to create a Student class, initialize student details using a parameterized constructor, and display the student information using class objects and member functions

```
class Student {  
    String name;  
    int age;  
    int rollNo;  
    Student(String n, int a, int r) {  
        name = n;  
        age = a;  
        rollNo = r;  
    }  
    void display() {  
        System.out.println("Name: " + name);  
        System.out.println("Age: " + age);  
        System.out.println("Roll No: " + rollNo);  
    }  
}  
public class Main {  
    public static void main(String[] args) {  
        System.out.println("The program is coded by Karunya Singh  
Kanyal, 2400320100580");  
        Student s1 = new Student("Ankit", 20, 101);  
        Student s2 = new Student("Rahul", 21, 102);  
        s1.display();  
        System.out.println();  
        s2.display();  
    }  
}
```

Output –

The program is coded by Karunya Singh Kanyal, 2400320100580

Name: Ankit

Age: 20

Roll No: 101

Name: Rahul

Age: 21

Roll No: 102

EXPERIMENT 3(b)

Question 6. Write a Java program to create a BankAccount class using encapsulation (private data members) and perform operations such as deposit, withdraw, and display balance using menu-driven choices.

```
import java.util.Scanner;
class BankAccount {
    private double balance;
    BankAccount(double initialBalance) {
        balance = initialBalance;
    }
    public void deposit(double amount) {
        if(amount > 0) {
            balance += amount;
            System.out.println("Amount deposited successfully");
        } else {
            System.out.println("Invalid amount");
        }
    }
    public void withdraw(double amount) {
        if(amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Amount withdrawn successfully");
        } else {
            System.out.println("Insufficient balance or invalid
amount");
        }
    }
    public void displayBalance() {
        System.out.println("Current Balance: " + balance);
    }
}
public class Main {
    public static void main(String[] args) {
        System.out.println("The program is coded by Karunya Singh
Kanyal, 2400320100580");
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter initial balance: ");
        double initialBalance = sc.nextDouble();
        BankAccount account = new BankAccount(initialBalance);
        int choice;
        do {
            System.out.println("\n1. Deposit");
            System.out.println("2. Withdraw");
            System.out.println("3. Display Balance");
```

```

System.out.println("4. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();

switch(choice) {
    case 1:
        System.out.print("Enter amount to deposit: ");
        double depositAmount = sc.nextDouble();
        account.deposit(depositAmount);
        break;

    case 2:
        System.out.print("Enter amount to withdraw: ");
        double withdrawAmount = sc.nextDouble();
        account.withdraw(withdrawAmount);
        break;

    case 3:
        account.displayBalance();
        break;

    case 4:
        System.out.println("Exiting...");
        break;

    default:
        System.out.println("Invalid choice");
}
} while(choice != 4);

sc.close();
}
}

```


Output –

The program is coded by Karunya Singh Kanyal, 2400320100580

Enter initial balance: 500000

1. Deposit
2. Withdraw
3. Display Balance
4. Exit

Enter your choice: 2

Enter amount to withdraw: 30000

Amount withdrawn successfully

1. Deposit
2. Withdraw
3. Display Balance
4. Exit

Enter your choice: 3

Current Balance: 470000.0

1. Deposit
2. Withdraw
3. Display Balance
4. Exit

Enter your choice: 4

Exiting...

EXPERIMENT 4(a)

Question 7. Write a Java program to demonstrate Compile-time Polymorphism (Method Overloading) by creating a class Calculator that performs addition of integers, floating values, and three numbers using overloaded methods.

```
class Calculator {
    int add(int a, int b) {
        return a + b;
    }

    float add(float a, float b) {
        return a + b;
    }

    int add(int a, int b, int c) {
        return a + b + c;
    }
}

public class Main {
    public static void main(String[] args) {
        System.out.println("The program is coded by Karunya Singh Kanyal, 2400320100580");

        Calculator calc = new Calculator();

        int result1 = calc.add(10, 20);
        float result2 = calc.add(5.5f, 4.5f);
        int result3 = calc.add(1, 2, 3);

        System.out.println("Addition of two integers: " + result1);
        System.out.println("Addition of two floats: " + result2);
        System.out.println("Addition of three integers: " + result3);
    }
}
```

Output –

The program is coded by Karunya Singh Kanyal, 2400320100580
Addition of two integers: 30
Addition of two floats: 10.0
Addition of three integers: 6

EXPERIMENT 4(b)

Question 8. Write a Java program to demonstrate Single Inheritance by creating a Person class and deriving a Student class from it, and display the details of the student using inherited properties and methods.

```
class Person {
    String name;
    int age;
    void setPersonDetails(String n, int a) {
        name = n;
        age = a;
    }
    void displayPerson() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
}
class Student extends Person {
    int rollNo;
    String course;
    void setStudentDetails(int r, String c) {
        rollNo = r;
        course = c;
    }
    void displayStudent() {
        displayPerson();
        System.out.println("Roll No: " + rollNo);
        System.out.println("Course: " + course);
    }
}
public class Main {
    public static void main(String[] args) {
        System.out.println("The program is coded by Karunya Singh Kanyal, 2400320100580");
        Student s = new Student();
        s.setPersonDetails("Ankit", 20);
        s.setStudentDetails(101, "CSE");
        s.displayStudent();
    }
}
```

Output –

The program is coded by Karunya Singh Kanyal, 2400320100580
Name: Karunya
Age: 19 Roll No: 101 Course: CSE

EXPERIMENT 4(c)

Question 9. Write a Java program to demonstrate Runtime Polymorphism (Method Overriding) by creating a base class Shape and derived classes Circle and Rectangle to override the area() method and display the area of each shape

```
import java.util.Scanner;
class Shape {
    void area() {
        System.out.println("Area not defined");
    }
}
class Circle extends Shape {
    double radius;

    Circle(double r) {
        radius = r;
    }
    void area() {
        double result = Math.PI * radius * radius;
        System.out.println("Area of Circle: " + result);
    }
}
class Rectangle extends Shape {
    double length, breadth;

    Rectangle(double l, double b) {
        length = l;
        breadth = b;
    }
    void area() {
        double result = length * breadth;
        System.out.println("Area of Rectangle: " + result);
    }
}

public class Main {
    public static void main(String[] args) {
        System.out.println("The program is coded by Karunya Singh Kanyal, 2400320100580");
        Scanner sc = new Scanner(System.in);
        Shape s;
        System.out.println("1. Circle");
        System.out.println("2. Rectangle");
        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();
    }
}
```

```

switch(choice) {
    case 1:
        System.out.print("Enter radius: ");
        double r = sc.nextDouble();
        s = new Circle(r);
        s.area();
        break;

    case 2:
        System.out.print("Enter length: ");
        double l = sc.nextDouble();
        System.out.print("Enter breadth: ");
        double b = sc.nextDouble();
        s = new Rectangle(l, b);
        s.area();
        break;

    default:
        System.out.println("Invalid choice");
}
sc.close();
}
}

```

Output –

The program is coded by Karunya Singh Kanyal, 2400320100580

1. Circle

2. Rectangle

Enter your choice: 1

Enter radius: 6

Area of Circle: 113.09733552923255

EXPERIMENT 4(d)

Question 10. Write a Java program to justify that Java does not support multiple inheritance using classes, but multiple inheritance can be achieved using interfaces by implementing more than one interface in a single class.

```
interface A {  
    void showA();  
}  
interface B {  
    void showB();  
}  
class C implements A, B {  
    public void showA() {  
        System.out.println("Method from Interface A");  
    }  
  
    public void showB() {  
        System.out.println("Method from Interface B");  
    }  
    void display() {  
        System.out.println("Class C implementing multiple  
interfaces");  
    }  
}  
public class Main {  
    public static void main(String[] args) {  
        System.out.println("The program is coded by Karunya Singh  
Kanyal, 2400320100580");  
        C obj = new C();  
        obj.showA();  
        obj.showB();  
        obj.display();  
    }  
}
```

Output -

```
The program is coded by Karunya Singh Kanyal, 2400320100580  
Method from Interface A  
Method from Interface B  
Class C implementing multiple interfaces
```